Statistical Business Analysis Report

Week 7 project - sales performance and customer churn

Project Overview

This report covers a statistical analysis of two datasets - a 100-row sales log and a 500-row customer churn extract. The goal is to work through descriptive statistics, distribution checks, correlation, hypothesis testing, confidence intervals and regression, and to end with a short set of business recommendations that the numbers actually support.

Data and Setup

business_data.csv holds 100 sales records with Date, Product, Quantity, Price, Customer_ID, Region and Total_Sales. customer_churn.csv holds 500 customer records with Tenure, MonthlyCharges, TotalCharges, Contract type, Payment method, billing preference, senior citizen flag and a Churn indicator. Both files were loaded with pandas and used as-is, no rows were dropped.

Setup: install the packages in requirements.txt with pip install -r requirements.txt, then open statistical_analysis.ipynb in Jupyter and run all cells in order. The notebook expects business_data.csv and customer_churn.csv in the same folder.

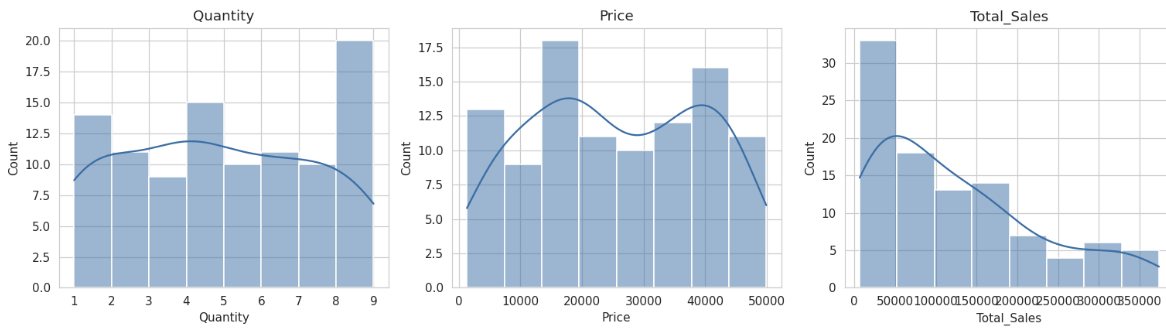
Descriptive Statistics

Average Total_Sales across the 100 orders is 123,650, with a standard deviation of 100,161 - close to the size of the mean itself, which points to a right-skewed distribution rather than a tight cluster around the average. Median sale is 97,956, noticeably below the mean, which confirms the skew. For the churn dataset, average tenure is 36.5 months and average monthly charge is 113.64, both fairly evenly spread with no extreme skew.

Metric	Quantity	Price	Total_Sales
Mean	4.78	25,808.51	123,650.48
Median	5.00	24,192.00	97,955.50
Std dev	2.59	13,917.63	100,161.09
Mode	4.00	1,308.00	6,540.00

Distribution Analysis

Histograms with a kernel density overlay were built for Quantity, Price and Total_Sales, then checked against a Shapiro-Wilk normality test.



All three metrics failed the Shapiro-Wilk test at the 0.05 level (Quantity: W = 0.930, $p < 0.001$; Price: W = 0.948, p = 0.001; Total_Sales: W = 0.899, $p < 0.001$), so none of them are normally distributed. This is the reason Welch's t-test (unequal variances) was used for the hypothesis tests below rather than a standard pooled-variance t-test.

Correlation Analysis

Quantity correlates with Total_Sales at r = 0.688 ($p < 0.001$) and Price correlates with Total_Sales at r = 0.646 ($p < 0.001$) - both strong and expected, since Total_Sales is the product of the two. In the churn dataset, Tenure and TotalCharges showed essentially no linear relationship (r = -0.006, p = 0.899), which is unusual and worth a data quality check before using TotalCharges in any billing analysis.

Hypothesis Testing

Five tests were run, summarized below. Alpha was set at 0.05 throughout.

#	Test	Comparison	Statistic	p-value	Result
1	Welch's t-test	Total_Sales, East vs West	t = 2.062	0.046	Significant
2	Welch's t-test	MonthlyCharges, churned vs retained	t = 2.646	0.010	Significant
3	Welch's t-test	Total_Sales, Laptop vs Phone	t = 0.565	0.575	Not significant
4	One-way ANOVA	Total_Sales across 4 regions	F = 2.164	0.097	Not significant
5	Chi-square	Contract type vs Churn	chi2 = 27.715	< 0.001	Significant

East region customers spend significantly more per order than West region customers (132,613 vs 81,689 average). Churned customers were paying more per month before leaving than customers who stayed (129.77 vs 111.72). There is no significant difference between Laptop and Phone sale values, and the four-region ANOVA does not reach significance overall, so the East/West gap should not be generalized into a broader regional trend without more data. Contract type is strongly associated with churn - full breakdown in the full test log, hypothesis_tests_results.txt.

Confidence Intervals

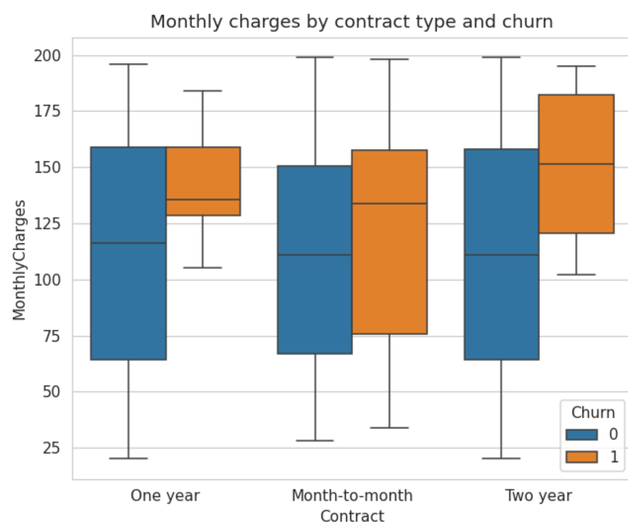
Metric	Estimate	95% CI
Average Total_Sales	123,650.48	(103,776.35, 143,524.61)
Churn rate	10.60%	(7.90%, 13.30%)
Average MonthlyCharges	113.64	(109.08, 118.19)

The Total_Sales confidence interval is wide, about plus or minus 19,874 (margin of error), which reflects the high variance already seen in the distribution analysis. A single-point average by itself is not a safe number to plan a forecast around here.

Regression Analysis

Model 1 - linear regression, Total_Sales ~ Quantity + Price. R-squared = 0.884, both predictors significant at p < 0.001. This mostly confirms the data is internally consistent, since Total_Sales is derived from these two fields.

Model 2 - logistic regression, Churn ~ Tenure + MonthlyCharges + SeniorCitizen. Pseudo R-squared = 0.618. Tenure has a negative, significant coefficient (p < 0.001) - each additional month of tenure lowers churn odds. MonthlyCharges has a small positive, significant coefficient (p = 0.003) - higher monthly bills push churn odds up slightly. SeniorCitizen status was not significant (p = 0.577).



Business Insights and Recommendations

1. West region sales are significantly below East - worth investigating pricing, product mix, or marketing reach in that region before assuming it is a demand problem.

2. Newer, higher-paying customers are the group most likely to churn - tenure strongly predicts retention and churned customers had higher monthly bills. A targeted retention check-in in the first few months looks like the highest leverage move.

3. Month-to-month contracts churn at a much higher rate than annual contracts - an incentive to move customers onto longer contracts should pay for itself in reduced churn.

4. Total_Sales figures should be read with their confidence interval in mind, not as a single fixed number, since the underlying distribution is wide and right-skewed.

5. The Tenure-TotalCharges relationship in the churn data does not add up (near-zero correlation) - flag this to whoever owns that data pipeline before billing decisions get made off it.

Methodology Notes

Welch's t-test was used instead of the standard Student's t-test wherever two group means were compared, since the normality checks in the distribution analysis section showed the sales metrics are not normally distributed and group sizes were not always equal. The one-way ANOVA and chi-square test use their standard assumptions; cell counts in the contract-by-churn table were all above 5, so the chi-square approximation is valid. All statistical work was done in Python with scipy.stats and statsmodels - full code is in statistical_analysis.ipynb.